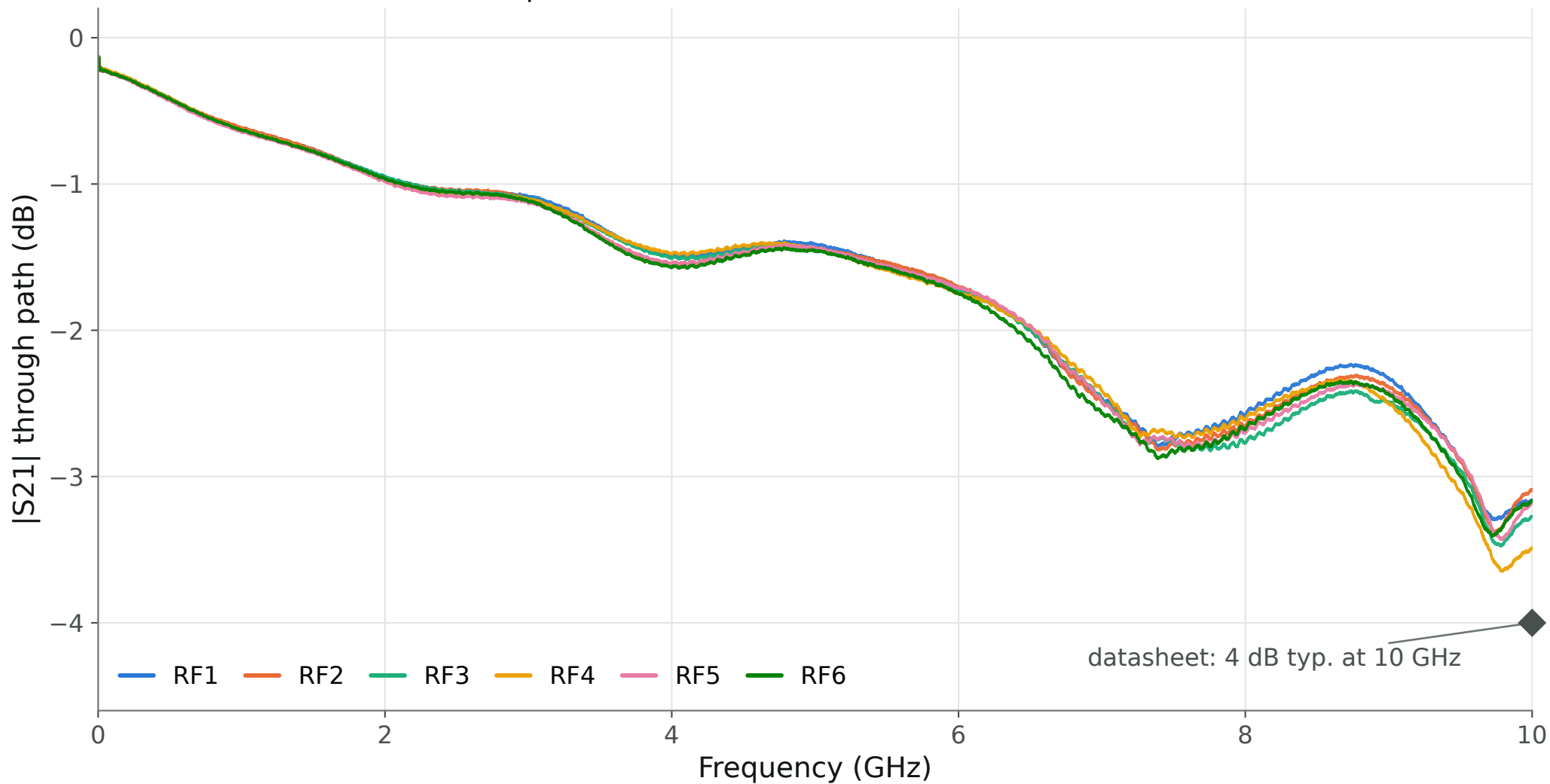
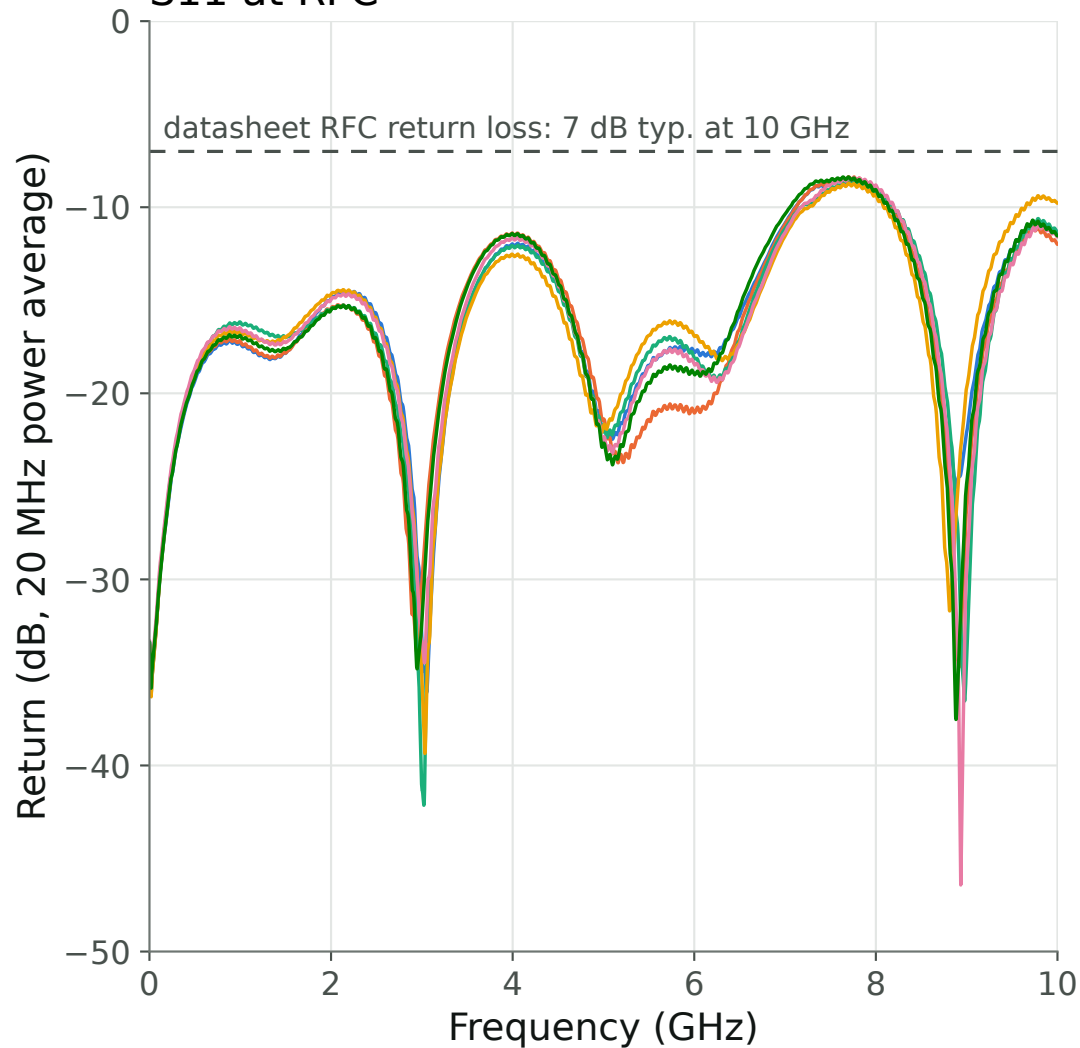


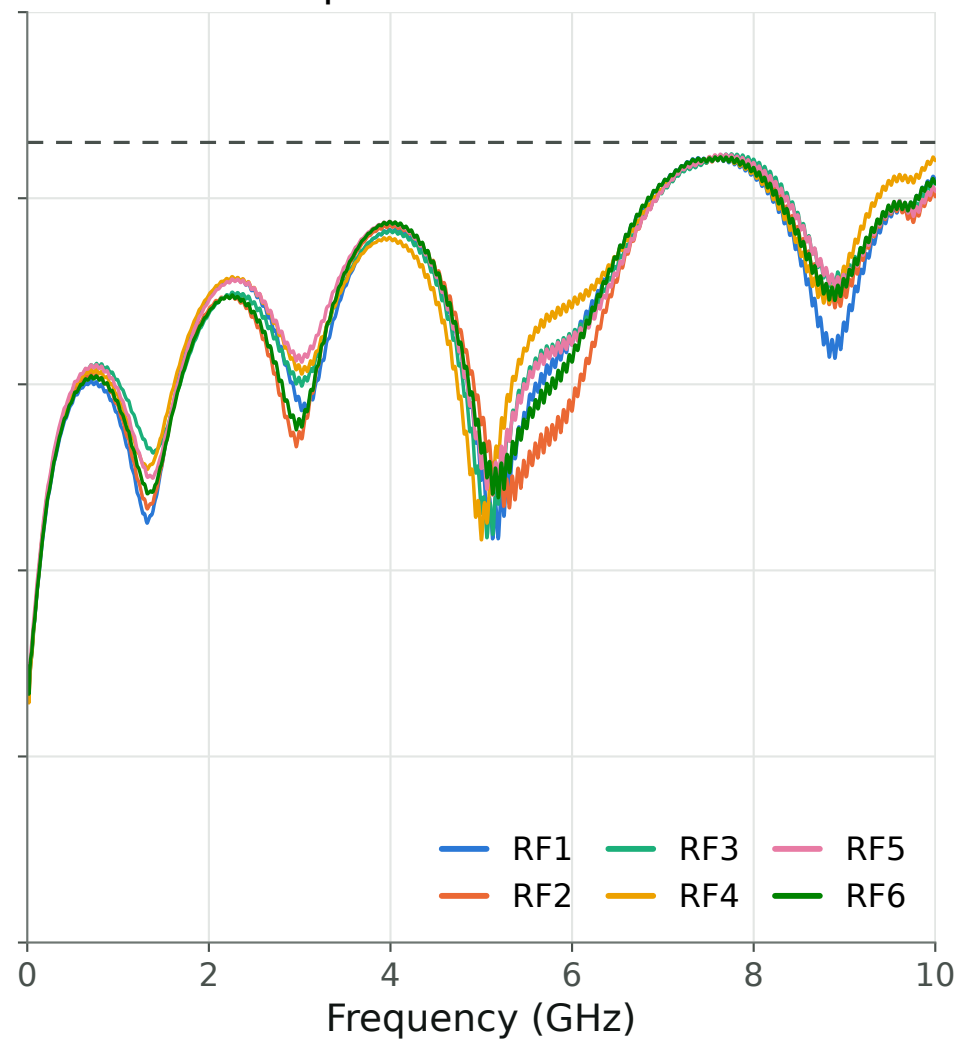
Insertion loss, RFC to each RF port (VNA-calibrated at the switch connectors)



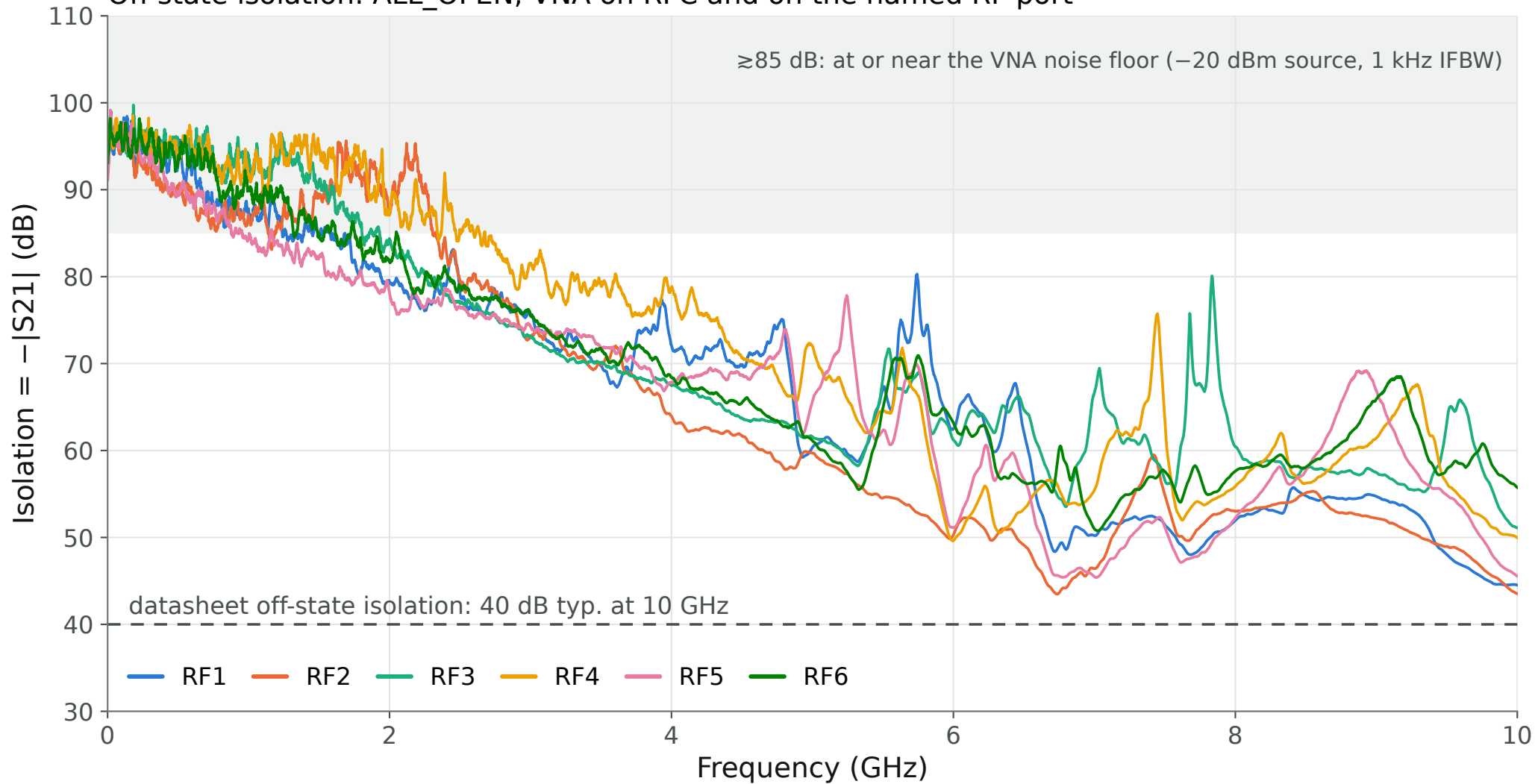
S11 at RFC



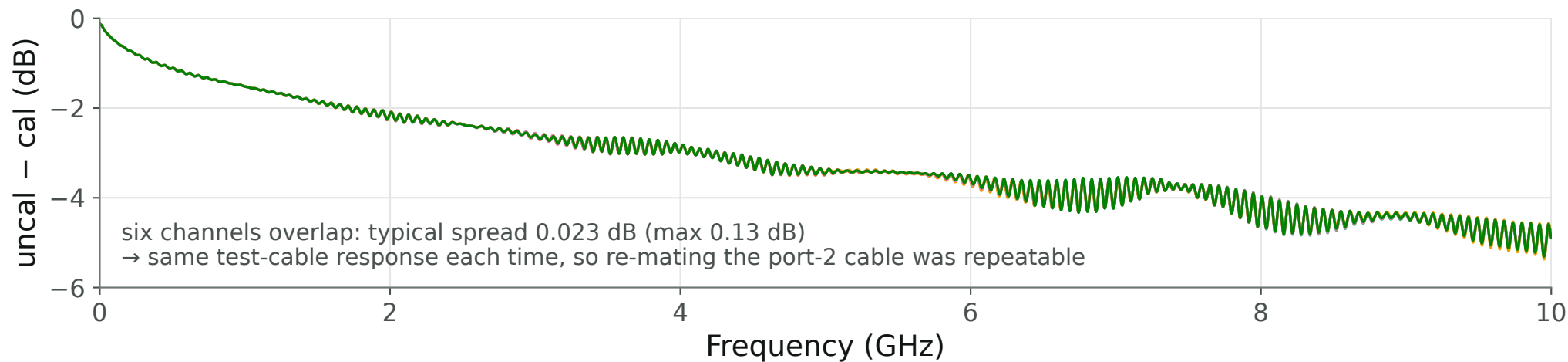
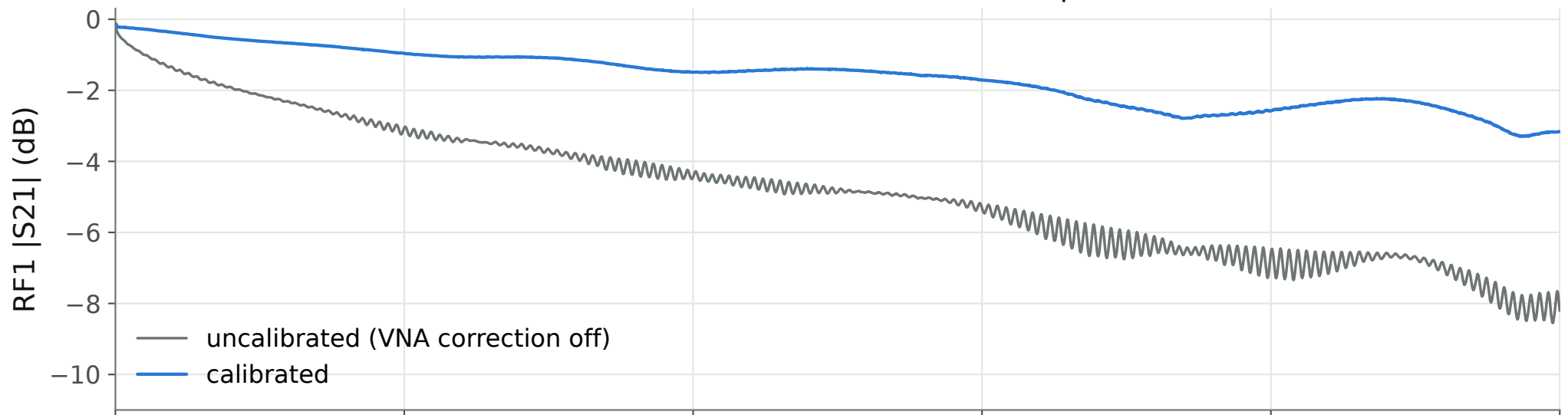
S22 at the RF port



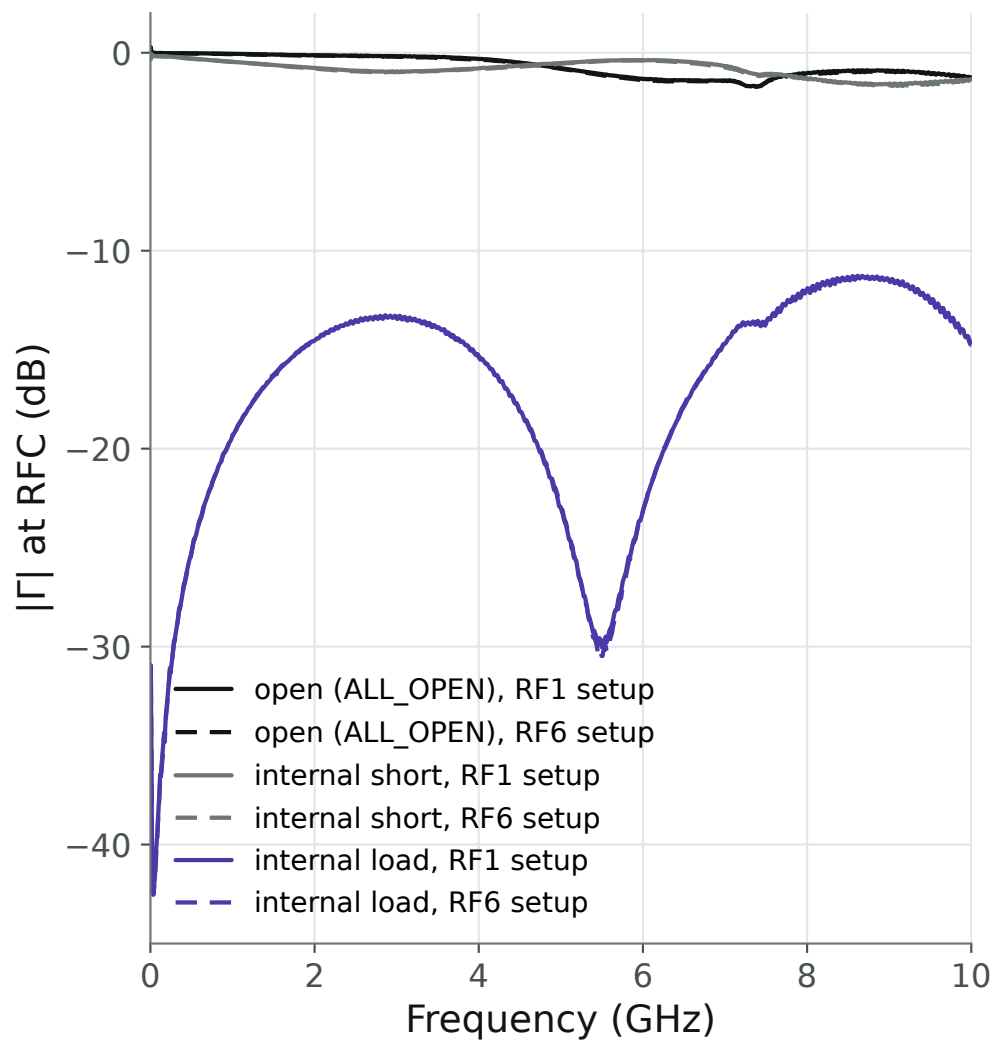
Off-state isolation: ALL_OPEN, VNA on RFC and on the named RF port



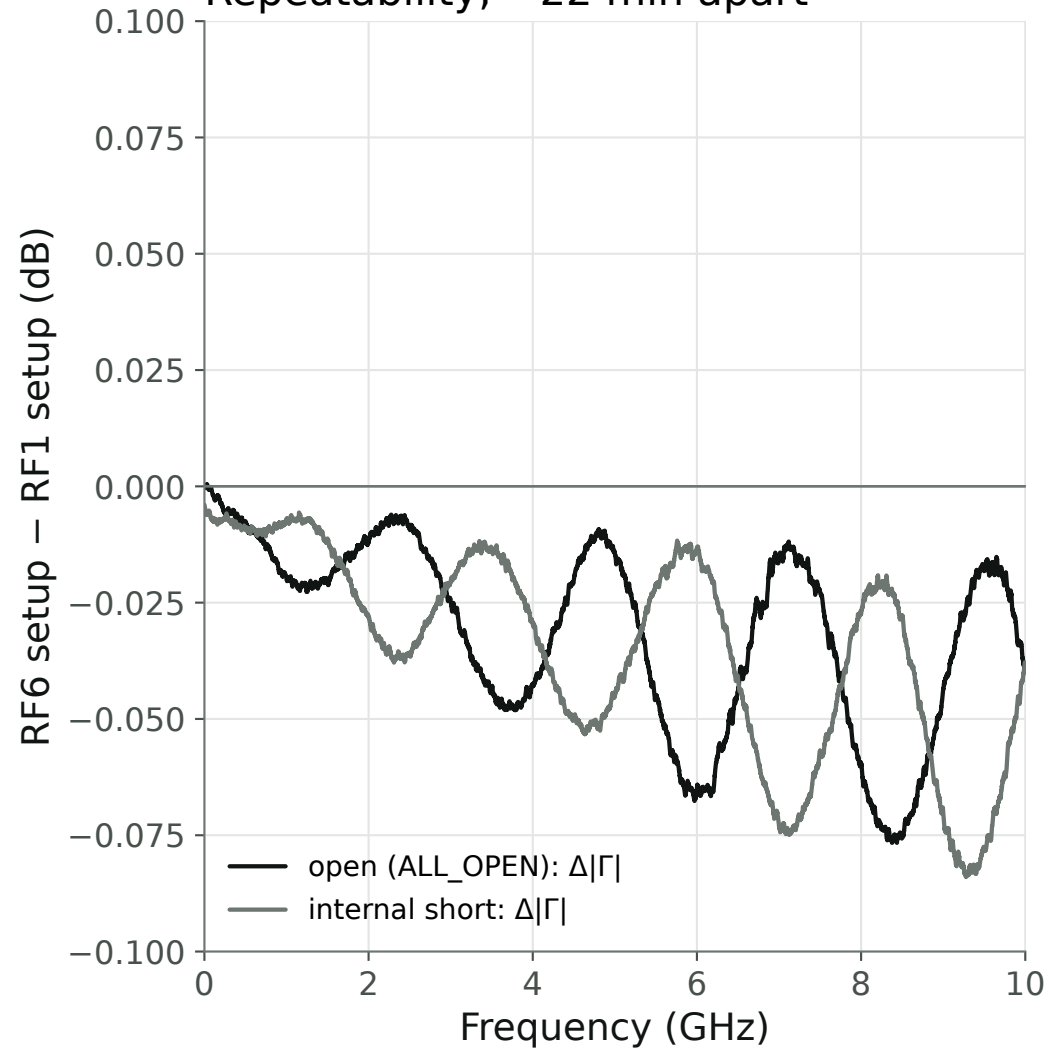
What the VNA calibration removes (test cables + VNA test-set response)



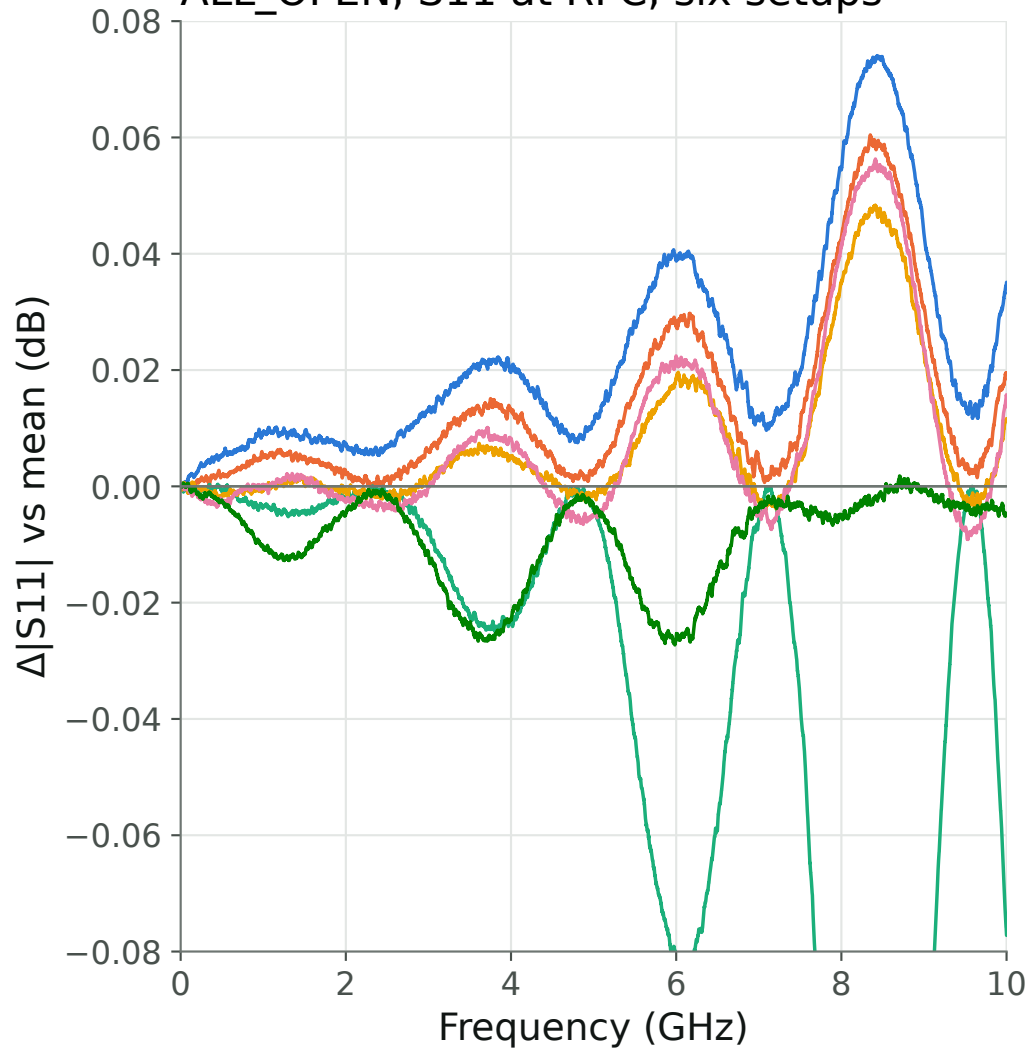
Internal standards seen at RFC



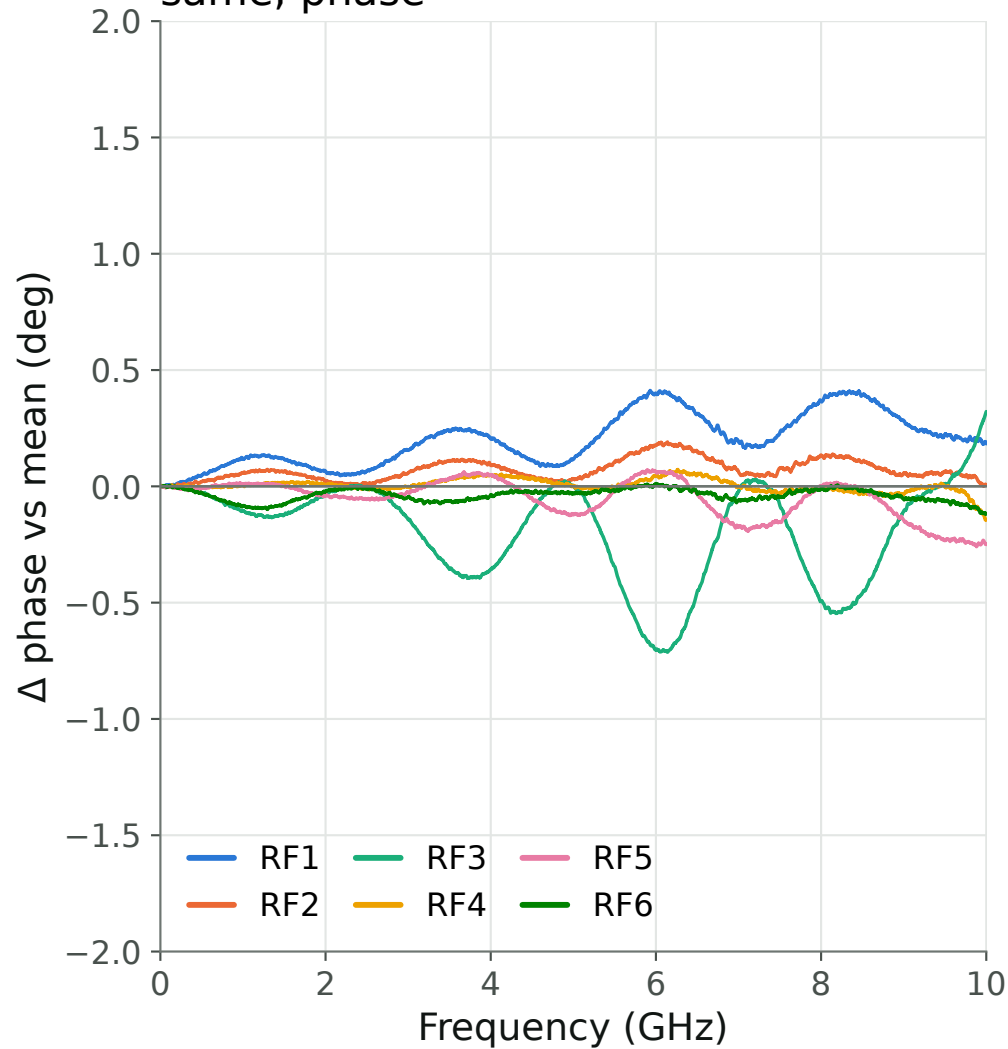
Repeatability, ~22 min apart



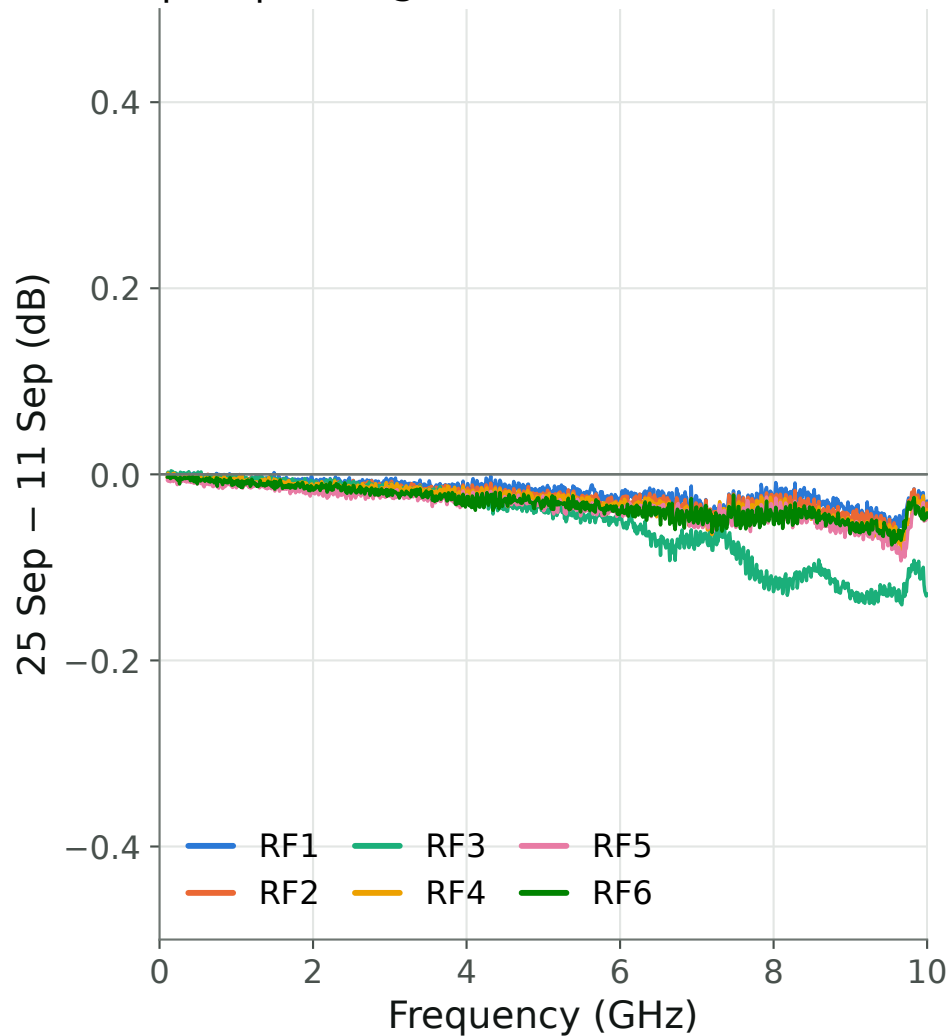
ALL_OPEN, S11 at RFC, six setups



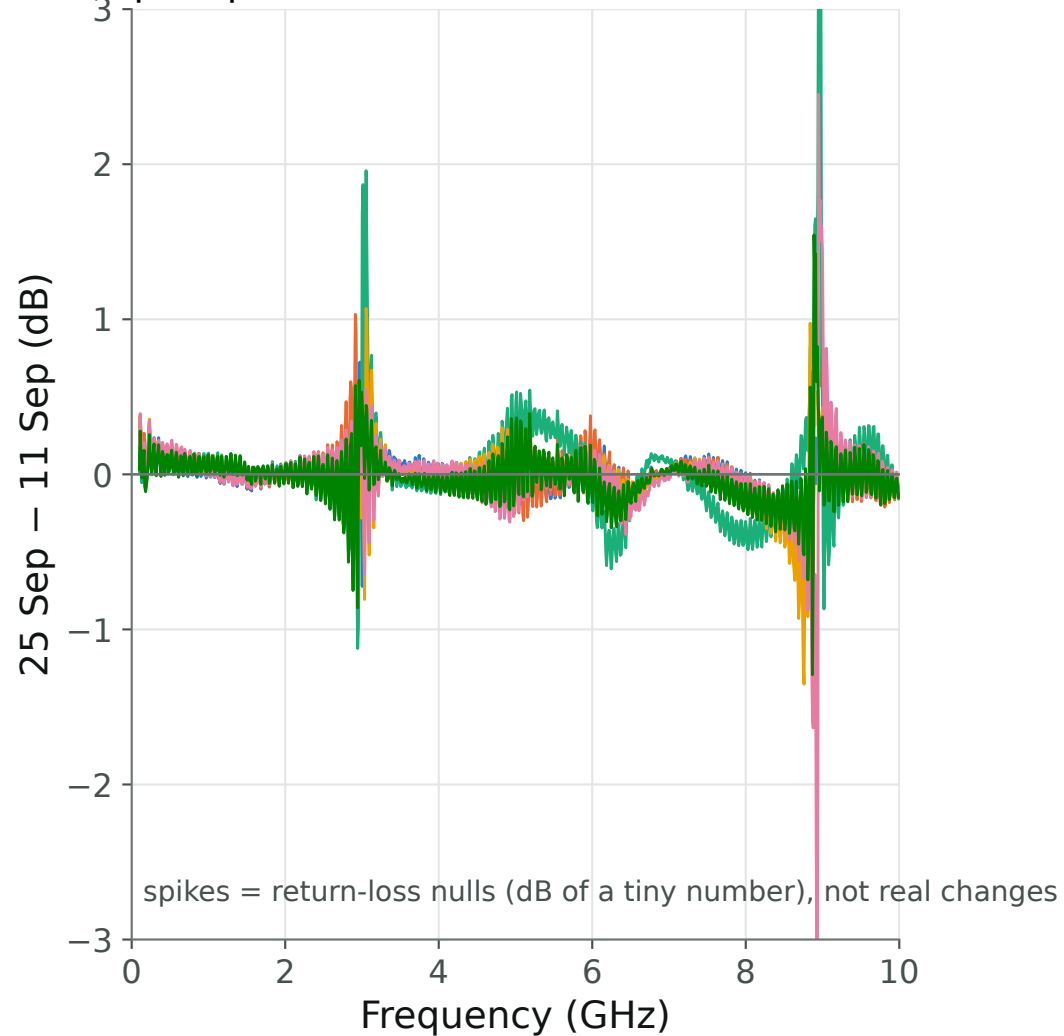
same, phase



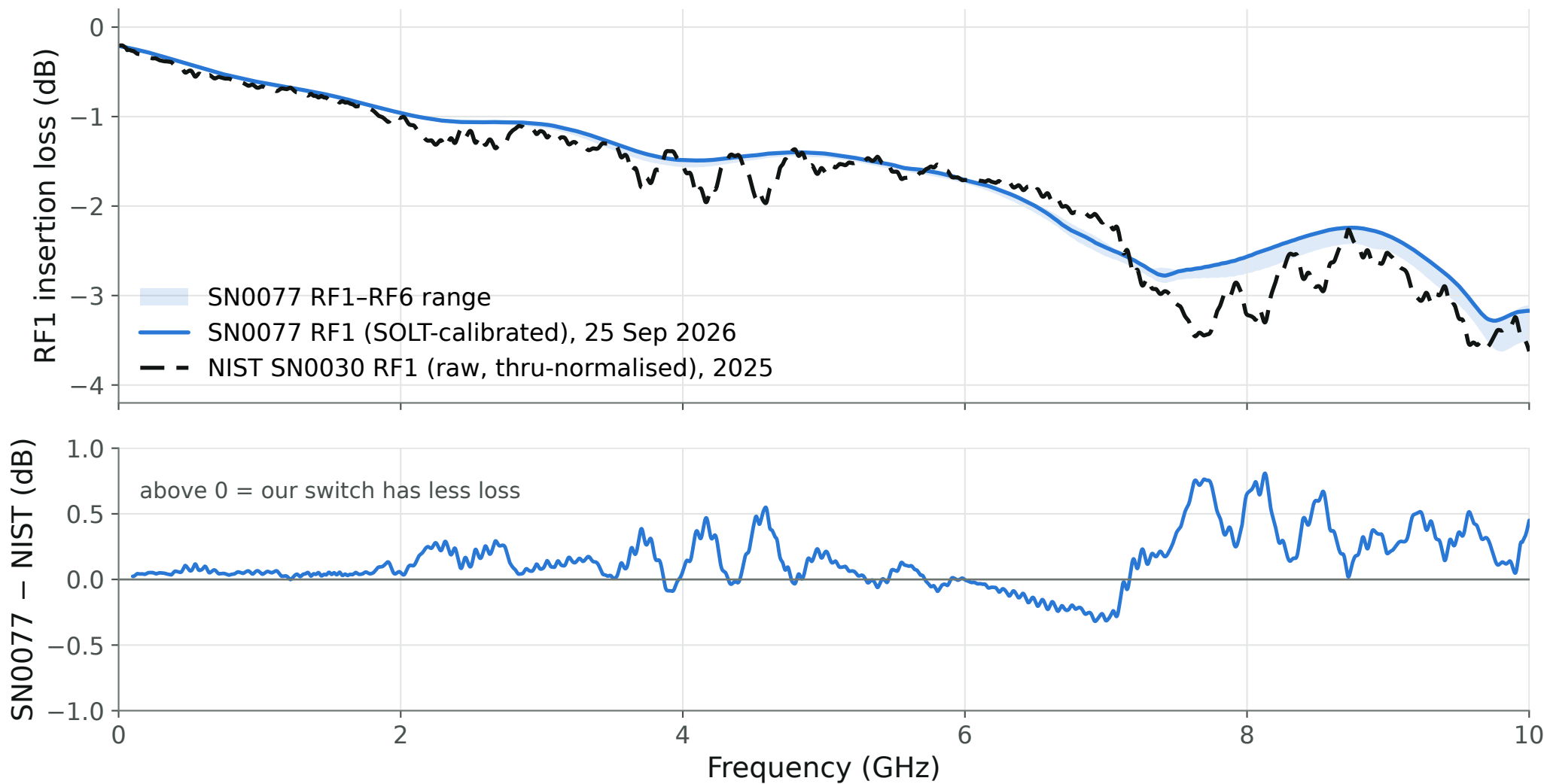
|S21| through



|S11| at RFC

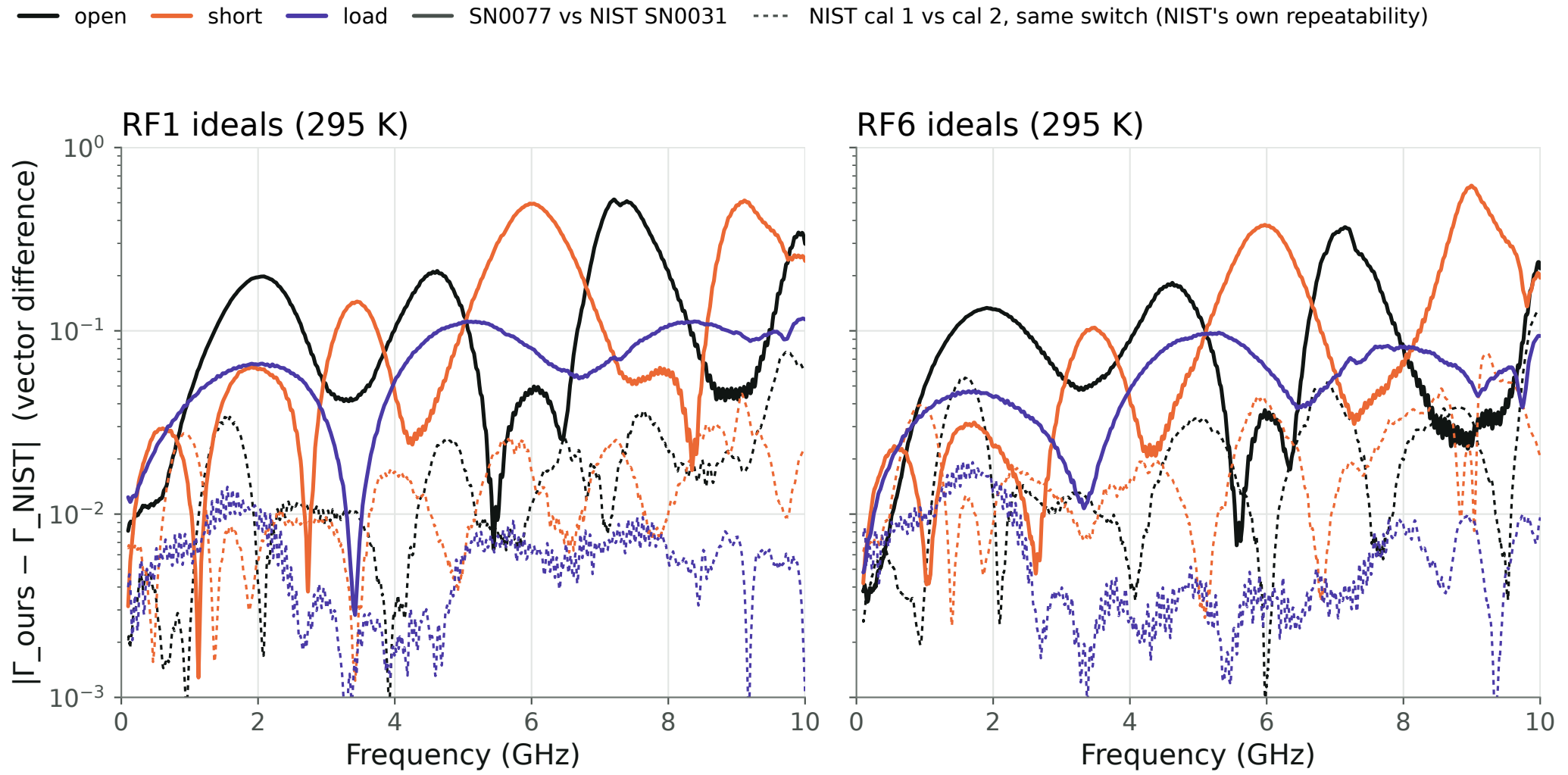


Insertion loss vs NIST at 295 K: both referenced to the switch SMA connectors



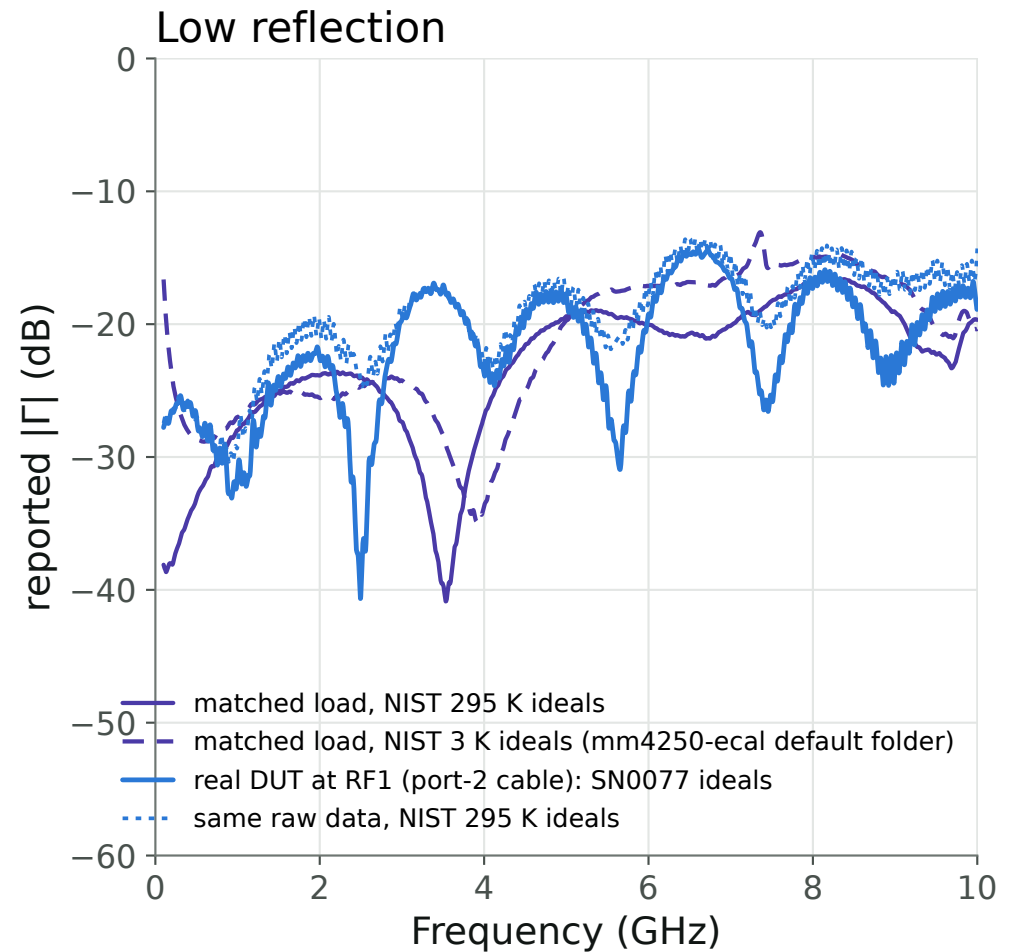
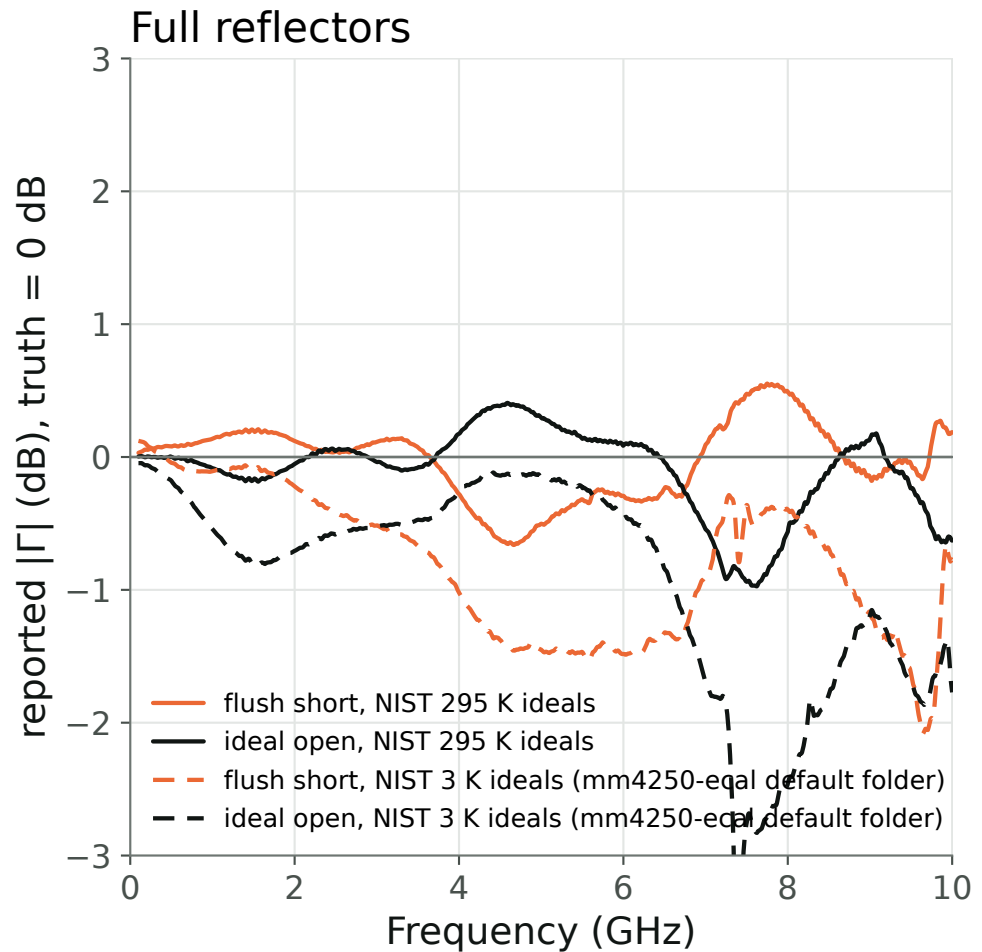
Both curves 100 MHz running mean (NIST's thru-normalisation leaves mismatch ripple a SOLT cal removes) · NIST thru repeats agree to 0.01 dB

Internal-standard definitions at the RF-port SMA plane: our SOLT-derived vs NIST's 295 K ideals



Ours: calibrated standard at RFC with the calibrated RFC→RFn path de-embedded · 50 MHz running mean · 0.01 $\approx$ -40 dB error vector

E-cal on SN0077 RF1 with NIST's generic ideals: the error you would get



Truth = SN0077 ideals derived from the SOLT data · 50 MHz running mean